Databricks Student Project Guide

Retail Data Warehouse + Generative AI Learning Path

Public GitHub: github.com/akhilvydyula01/databricks

Workspace: dbc-37e2a2e8-2739.cloud.databricks.com

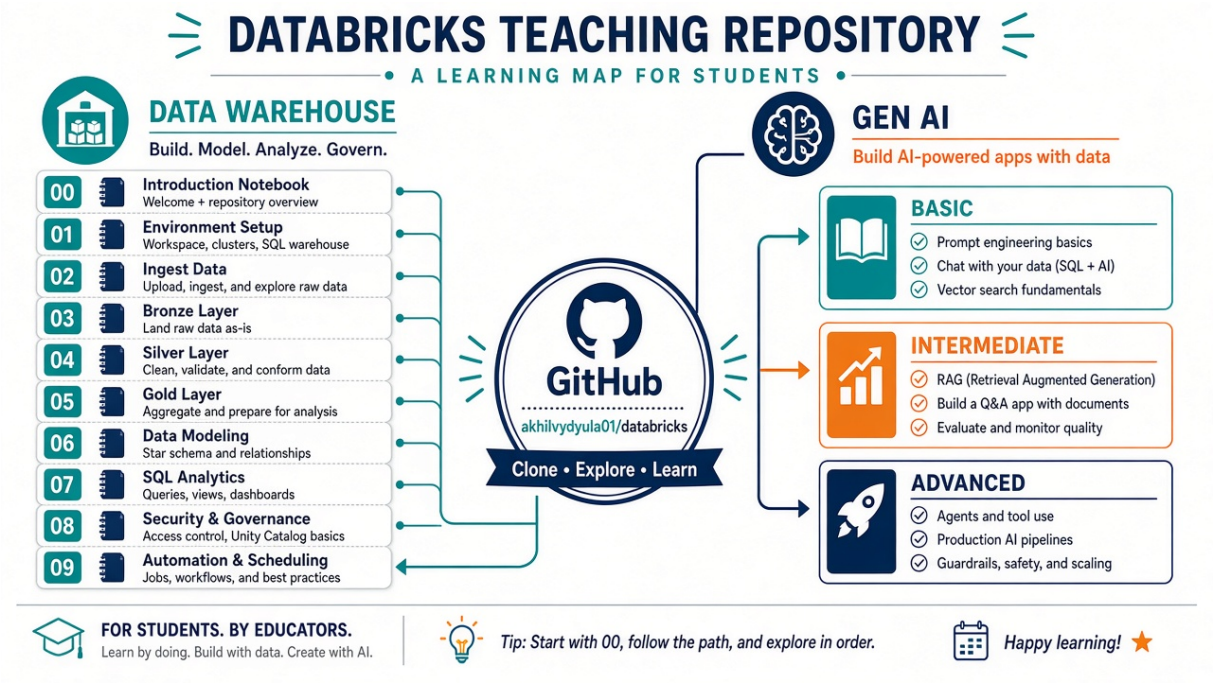


Figure 1. Full repository map — Data Warehouse track and Gen AI track.

This guide is designed for classroom teaching and YouTube walkthroughs. Students should clone the repository into Databricks Repos / Git folders and run notebooks in order.

1. Project Overview

This repository contains two teaching tracks that share one Databricks workspace and one public GitHub repo.

Track	Path	What students learn
Data Warehouse	notebooks/data_warehouse	Medallion architecture, star schema, SCD2, quality, SQL analytics
Gen AI Basic→Advanced	notebooks/gen_ai + gen-ai/	Open-source LLMs, RAG, agents, LoRA, multi-agent patterns
Teaching pack	data-warehouse/	YouTube scripts, assignment, job JSON, 20-min demo plan

1.1 Learning outcomes

- Explain lakehouse and medallion architecture in plain language.
- Build Bronze, Silver, and Gold tables for a retail domain (NovaMart).
- Design a star schema with facts and dimensions.
- Implement SCD Type 2 for changing customer attributes.
- Apply data quality gates before analytics.
- Optionally explore open-source Gen AI notebooks in the same repo.

2. How to Open the Project in Databricks

2.1 One-time setup

- Open your Databricks workspace.
- Go to Workspace → Repos / Git folders.
- Add repo URL: https://github.com/akhilvydyula01/databricks
- Click Pull to sync latest notebooks.
- Attach a Databricks Runtime 13+ cluster (ML runtime for Gen AI notebooks).

2.2 Exact notebook locations

Data Warehouse: Repos → <you> → databricks → notebooks → data_warehouse
Gen AI: Repos → <you> → databricks → notebooks → gen_ai → basic / intermediate / advanced

2.3 Suggested class schedule

Session	Focus	Notebooks
Day 1	Architecture + Bronze/Silver	00, 01, 02, 03
Day 2	Gold + SCD2 + Quality	04, 05, 06
Day 3	Analytics + Jobs + Assignment	07, 08/09 + homework
Optional	Gen AI intro	gen_ai/basic 01–03

3. Data Warehouse Architecture

NovaMart is a fictional retailer. Students ingest customers, products, stores, and orders, then transform them through Bronze → Silver → Gold.

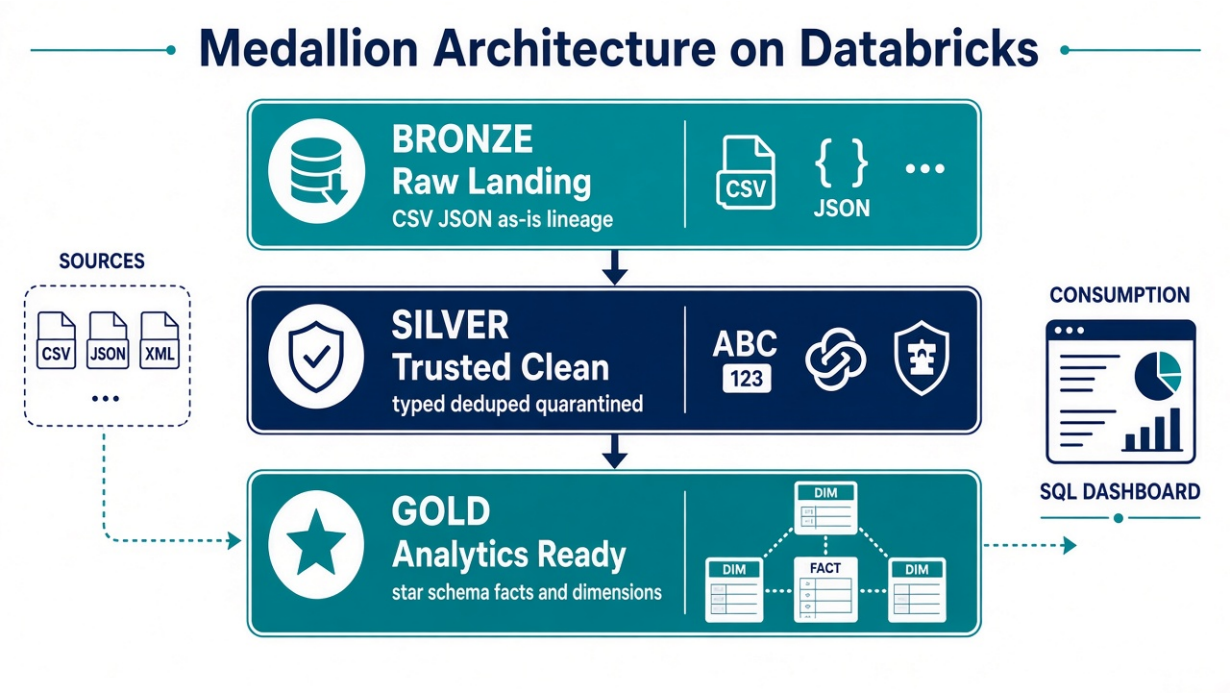


Figure 2. Medallion architecture used in this course.

3.1 Layer responsibilities

Layer	Purpose	Example tables
Bronze	Raw landing + lineage	bronze_orders, bronze_customers
Silver	Trusted cleaned data	silver_orders, silver_orders_rejected
Gold	BI-ready star schema	gold_fact_sales, gold_dim_*

Teaching tip: Emphasize that Bronze preserves history of what arrived. Silver creates trust. Gold creates usability for analysts.

4. Notebook Walkthrough (Data Warehouse)

#	Notebook	Student takeaway
00	architecture_overview	Big picture before coding
01	setup_and_sample_data	Create schema + landing files
02	bronze_ingestion	Land raw tables with metadata
03	silver_transform	Cast, dedupe, quarantine bad rows
04	gold_star_schema	Build dims + fact_sales
05	scd2_customers	Track customer history over time
06	data_quality	Fail pipeline on broken rules
07	analytics_sql	Answer business questions
08	end_to_end_pipeline	Run full refresh sequence
09	youtube_20min_runner	Fast demo path for teaching videos

4.1 Important demo details baked into sample data

- Order amounts arrive as strings in Bronze (type casting lesson).
- Duplicate order_id 5015 exists (dedupe lesson).
- One negative amount is rejected into silver_orders_rejected.
- Customer 101 later moves Austin → Houston (SCD2 lesson).

5. Gold Star Schema Explained

Analysts think in business events (facts) and descriptive context (dimensions). In this project, the fact grain is one order.

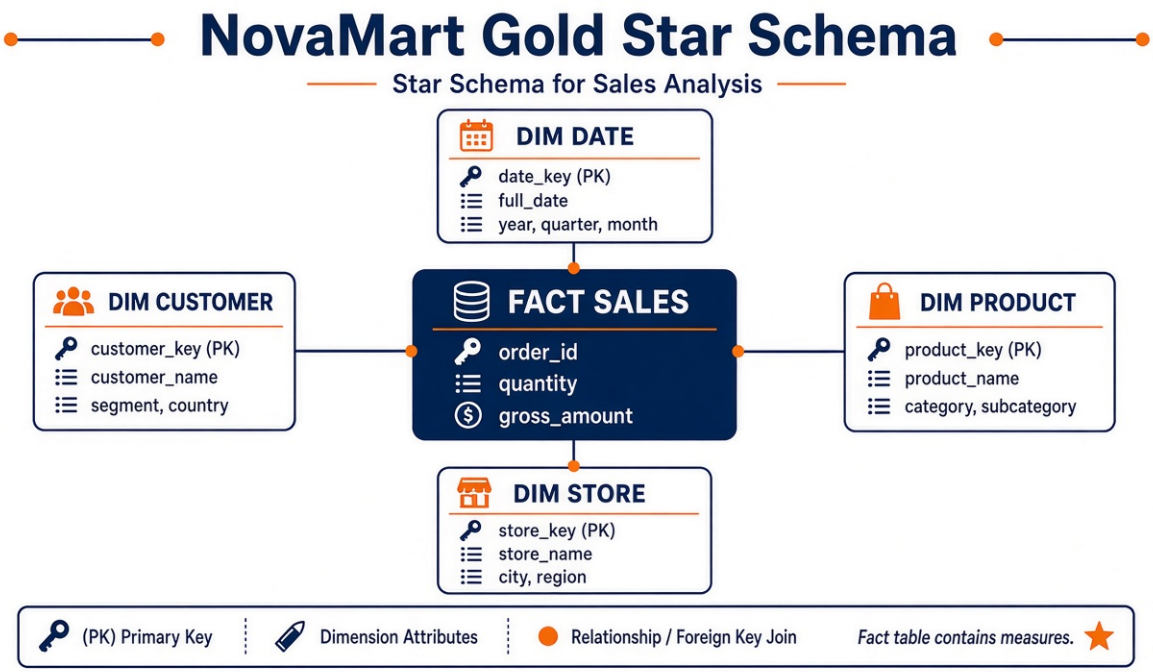


Figure 3. NovaMart Gold star schema.

5.1 Key definitions for exams / interviews

- **Fact table:** stores measures (quantity, gross_amount) and foreign keys.
- **Dimension table:** stores descriptive attributes (customer city, product category).
- **Surrogate key:** warehouse-generated key (customer_sk) independent of source IDs.
- **Grain:** what one fact row represents — here, one order.

6. SCD Type 2 (Customer History)

Slowly Changing Dimensions Type 2 preserve history when attributes change. Instead of overwriting Austin with Houston, we close the old row and insert a new current row.

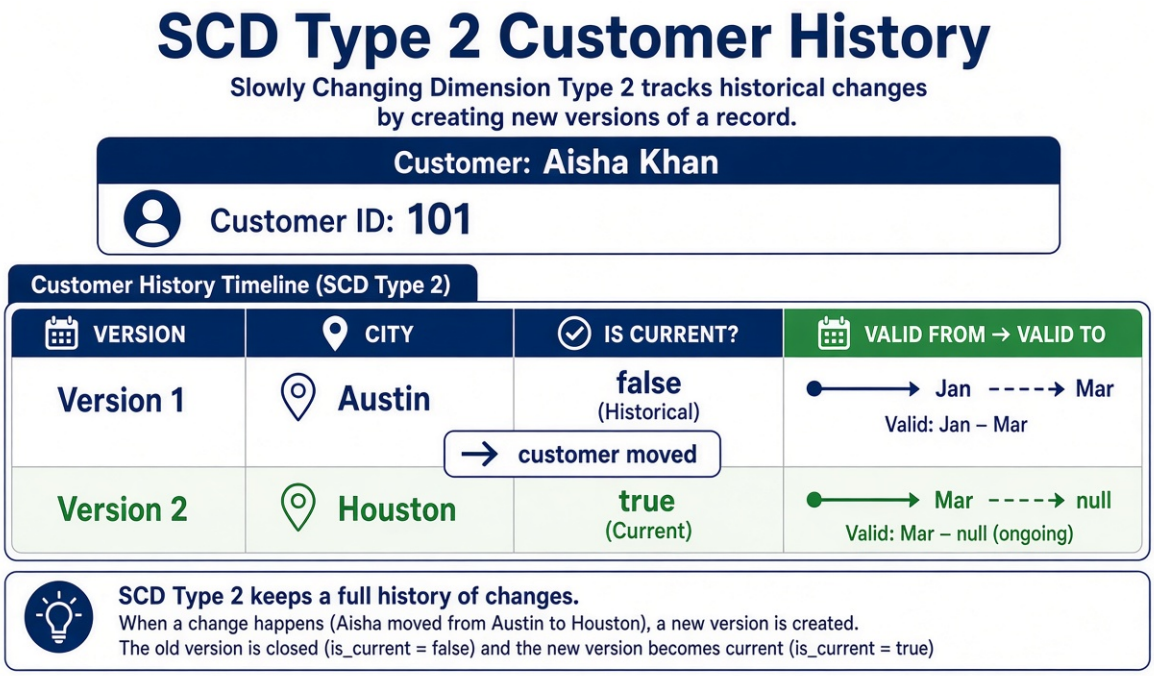


Figure 4. SCD Type 2 example for customer 101.

Columns to memorize: **valid_from**, **valid_to**, **is_current**. Advanced designs join the dimension version that was valid on the order date.

7. Data Quality and Analytics

7.1 Quality gates in notebook 06

- Fact sales_key must be unique.
- Quantity and gross_amount must be positive.
- No orphan keys to customer/product/store dimensions.
- Reject table must capture known bad records.

7.2 Business questions in notebook 07

- Total revenue and average order value
- Revenue by month
- Top products by revenue
- Store performance and AOV
- Customer segment contribution

7.3 Student assignment highlight

Homework asks students to add a **Returns** source and build bronze_returns → silver_returns → gold_fact_returns, then write SQL for refunds by product category. Full rubric: data-warehouse/STUDENT_ASSIGNMENT.md

8. Gen AI Track (Same Repository)

The same repo includes an open-source Gen AI path so students can continue into LLM apps without leaving Databricks.

Level	Folder	Projects
Basic	gen_ai/basic	Hello LLM, chatbot, tiny RAG
Intermediate	gen_ai/intermediate	FAISS RAG, tool agent, evaluation
Advanced	gen_ai/advanced	LoRA fine-tune, multi-agent, production RAG

Local Python equivalents live under **gen-ai/** with requirements.txt for offline practice. Databricks notebook versions are under **notebooks/gen_ai/**.

9. Quick Glossary

Term	Meaning
Lakehouse	Platform combining data lake storage with warehouse-style analytics
Medallion	Layered data design: Bronze, Silver, Gold
Star schema	Fact table surrounded by dimensions
SCD2	Keep historical versions of dimension attributes
Lineage	Tracking where data came from and when it was ingested
Quarantine	Isolating bad records instead of deleting silently

10. Resources

- GitHub repo: https://github.com/akhilvydyula01/databricks
- DWH docs: data-warehouse/README.md
- YouTube full script: data-warehouse/YOUTUBE_TEACHING_GUIDE.md
- 20-min demo cut: data-warehouse/YOUTUBE_20MIN_DEMO.md
- Voiceover teleprompter: data-warehouse/youtube/
- Job setup: data-warehouse/jobs/HOW_TO_CREATE_JOB.md

Instructor note: Print this PDF for students, or share the GitHub docs folder. For live class, project Figures 2–4 while running notebooks 02, 04, and 05.